A 12 V Hot-Swap / eFuse Front-End Using the LTC4215

Electronic Engineering Technical Documentation

Project Logbook

Version 0.1 — October 27, 2026

Abstract

This document describes the design, LTspice validation, and KiCad schematic/PCB layout of a 12 V hot-swap / eFuse front-end built around the LTC4215 hot-swap controller. It is a deliberately small project in the same series as the LM317, LM2596, and LTC1929 VRM projects – chosen specifically to add a fast, self-contained result to the portfolio without derailing the in-progress 2-phase VRM work, while still being directly on-theme for a server/GPU power-delivery specialization (live board insertion, inrush limiting, and electronic circuit breaking are the Phase 3 "server-specific power topics" item on that roadmap). Documented as an honest engineering logbook: every result is marked as **calculated**, **simulated**, or **pending** – never filled in with invented placeholder data. **Key result**: the controlled startup, the automatic recovery from an undervoltage (UV) event, and the permanent latch-off on a hard output short were all reproduced and numerically verified from the LTspice `.raw` file (not read visually from plots), matching the design's calculated circuit-breaker threshold (5 A) and confirming the LTC4215's documented UV auto-retry and OC latch-off behavior. The KiCad schematic and 2-layer PCB layout are complete, ERC/DRC clean, and not yet sent to fabrication (simulation + layout only, confirmed project scope).

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1 Project Objective

Design and validate, in LTspice, a 12 V hot-swap / electronic-fuse (eFuse) front-end using a real hot-swap controller IC (LTC4215): a MOSFET series pass switch whose gate is ramped under closed-loop control at insertion, with input undervoltage/overvoltage lockout and a fast, non-resettable circuit-breaker response to a downstream short. Scope agreed for this project: LTspice simulation first (this document), KiCad schematic/PCB afterward – not a build.

2 Controller Selection

2.1 Why hot-swap/eFuse, and why now

This project reaches ahead to the "server-specific power topics" item of the longer-term roadmap (hot-swap/eFuse, power sequencing, digital power/PMBus) rather than following the roadmap strictly in order, deliberately chosen as a small, fast, self-contained result while the 2-phase VRM (`VRM_2PHASE_LTC1929`) layout is in progress. It stays on-theme for the stated specialization (server/GPU board power delivery) instead of diluting the portfolio with an unrelated topic.

2.2 IC selection: LTC4215

Confirmed present in the local LTspice installation: SPICE subcircuit model (`LTC4215.sub`) and schematic symbol (`PowerProducts/LTC4215.asy`), both verified directly against the real datasheet pinout before use – not assumed. LTC4215 is a hot-swap controller with an external N-channel pass MOSFET, adjustable foldback current limiting, a programmable soft-start (dI/dt) ramp, input UV/OV lockout, and an independent, fast circuit-breaker comparator – a real part used in "Computers, Servers" and "Platform Management" applications per its own datasheet.

2.3 Package decision: QFN-24 vs. SSOP-16

The datasheet's simplified pinout comparison revealed a real functional difference between packages, not just a footprint choice:

	SSOP-16 (GN)	**QFN-24 (UFD)**
OV pin	absent	present
$\overline{\text{EN}}$ pin	absent	present
SENSE^+	tied internally to V_{DD}	separate external pin
ADIN, ADR1, ADR2	absent	present

This design uses the OV protection pin, so the **QFN-24 package is required** (the SSOP-16 cannot implement the OV divider at all). The LTspice macromodel exposes the full pin set regardless of package, so the simulation is package-agnostic.

Resolved (Section 6): symbol/footprint/3D model for `LTC4215IUFD-1#PBF` (SnapEDA, via DigiKey) – a real, in-production part confirmed against its own datasheet (`421513fc.pdf`, Rev. C) to be electrically equivalent to the LTC4215 modeled in Section 2.2 for every feature this project uses.

3 Datasheet Review

Transcribed directly from the LTC4215/LTC4215-2 datasheet (Linear Technology/Analog Devices, Rev. E), `docs/LTC4215_Datasheet.pdf`.

Table 1: LTC4215 key parameters

Parameter	Value	p.
Supply voltage range, V_{DD}	2.9–15 V	3
V_{DD} undervoltage lockout (rising)	2.84 V typ	3
V_{DD} overvoltage lockout	15.6 V typ	3
UV/OV/ON/FB comparator threshold	1.235 V typ (all four)	4
Circuit breaker threshold, $\Delta V_{SENSE(TH)}$	25 mV typ	3
OC fault filter, $t_{D(OC)}$ (LTC4215)	20 µs typ	3
Steady-state current limit, ΔV_{SENSE} (post-startup)	75 mV typ	3
External gate drive, ΔV_{GATE}	5.9 V typ (4.7–6.5)	3
Package	16-lead SSOP or 24-pin QFN	2

SS pin: sets the inrush dI/dt. Datasheet's own calibration point: 68 nF on SS gives a 5 mV ms^{-1} sense-voltage ramp (p.9), i.e. 1 A ms^{-1} with $R_{SENSE} = 5\,\text{m}\Omega$.

TIMER pin: normally a capacitor sets a 12.3 ms µF^{-1} start-up watchdog duration. **Documented alternate mode, used here**: "An internal timer provides a 100 ms start-up time and 5 s auto-retry time if this pin is tied to INTV$_{CC}$" (p.9) – simpler than computing a custom RC value, and 100 ms is ample margin over this design's calculated charge time (Section 5.3).

ON pin: "if the ON pin is tied high, then the FET On bit goes high 100 ms after power-up" (p.8) – the same 100 ms delay as above, and critical to the simulation debugging in Section 5.2.

4 Design

4.1 Design targets

V_{DD}	12 V (project-wide rail)
Circuit breaker trip	5 A (fast, non-resettable: *ocautoretry* = 0)
Steady-state current limit	15 A (post-startup foldback release)

4.2 Reference design: adapted from Datasheet Figure 1

Figure 1 ("12V Application with 5A Circuit Breaker") was adopted as the base reference – a real, manufacturer-validated 12 V application circuit, directly matching this project's target rail and current class. **System-level simplification:** Figure 1 shows two mating connectors (Connector 2 = backplane socket, Connector 1 = plug-in card edge, carrying both the power pins and the I^{2}C signaling pins to a backplane management controller). Without a backplane chassis to build, this project collapses both to a single input connector carrying only the 12 V/GND power pair; the I^{2}C signaling pins (SDA, SCL, $\overline{\text{ALERT}}$, ADR0–2, ADIN, GPIO) are left unused (Section 4.2.1).

4.2.1 I²C scope decision

This project uses only the LTC4215's analog hot-swap functionality (soft-start, UV/OV, current limit, circuit breaker) – no I²C monitoring. Two independent reasons: (1) keeping the project small and self-contained, matching its role as a fast portfolio addition; (2) the local LTspice macromodel does not even expose SDA/SCL/$\overline{\text{ALERT}}$/ADR0–2/ADIN as separate pins (confirmed from the `.asy` pin list), so I²C could not be simulated here regardless.

PENDING

A follow-on project (already on the longer-term roadmap, Phase 3) is to add an STM32 minimal system reading the LTC4215's I²C telemetry (8-bit ADC monitoring current/voltage) – the natural bridge between this project's power-electronics work and embedded-systems work. Not started.

4.2.2 Component values

Component	Value	Source / derivation
R_{SENSE} (R6)	5 mΩ	matches datasheet's own SS calibration point
M_1	Si7336ADP	see Section 4.2.3
D_1 (TVS, real BOM)	P6KE16A	Fig. 1 (surge suppressor, V_{DD}–GND)
D_1 (SPICE substitute)	BZX84C15L	see Section 4.2.4
$R_5/R_4/R_3$ (UV/OV string)	34.8 kΩ / 1.18 kΩ / 3.4 kΩ	Fig. 1, verified by calculation below
C_3	0.1 µF	Fig. 5 (noise bypass at UV/OV node)
R_7/R_8 (FB divider)	30.1 kΩ / 3.57 kΩ	Fig. 1
C_1 (SS)	68 nF	Fig. 1; sets 1 A ms^{-1} inrush ramp
TIMER	tied to INTV$_{CC}$	datasheet alternate mode (Section 3)
ON	tied to INTV$_{CC}$	auto-start at power-up
$\overline{\text{EN}}$	tied to GND	board always present (no mechanical detect)
R_1 (gate)	10 Ω	series gate-drive damping
R_2 (GPIO pull-up)	24 kΩ to INTV$_{CC}$	Fig. 1; power-good flag, kept even without I²C
C_4 (C_L, output)	12 mF	datasheet's own Start-Up Waveform test condition

UV/OV string, verified by calculation. A single resistor string from V_{DD} to GND ($R_5 = 34.8\,\text{k}\Omega$, $R_4 = 1.18\,\text{k}\Omega$, $R_3 = 3.4\,\text{k}\Omega$) feeds both comparators: the UV tap sits between R_5 and R_4, the OV tap between R_4 and R_3. Total $R_5 + R_4 + R_3 = 39.38\,\text{k}\Omega$. With both comparator thresholds at 1.235 V:

$$V_{DD,\,UV\,trip} = 1.235 \times \frac{39.38}{R_4 + R_3} = 1.235 \times \frac{39.38}{4.58} = \boxed{10.62\,\text{V}}\ \text{(falling)} \tag{1}$$

$$V_{DD,\,OV\,trip} = 1.235 \times \frac{39.38}{R_3} = 1.235 \times \frac{39.38}{3.4} = \boxed{14.30\,\text{V}}\ \text{(rising)} \tag{2}$$

i.e. roughly −12%/+19% around the 12 V nominal rail – sensible UV/OV margins, confirming the transcribed resistor values are self-consistent and not misread.

FB (power-good) divider, verified similarly: $R_7 = 30.1\,\text{k}\Omega$, $R_8 = 3.57\,\text{k}\Omega$, total $33.67\,\text{k}\Omega$:

$$V_{OUT,\,power\,bad} = 1.235 \times \frac{33.67}{3.57} = \boxed{11.65\,\text{V}} \tag{3}$$

power-bad is flagged below ≈ 97% of nominal – a typical margin.

4.2.3 MOSFET

Figure 1's real BOM choice (FDC653N) has no SPICE model in the local LTspice installation. Rather than searching for a new part, this project reuses **Si7336ADP** (Vishay PowerPAK SO-8) – already verified with a real SPICE model and a hand-built, datasheet-checked KiCad footprint in the `VRM_2PHASE_LTC1929` project. $V_{DS} = 30\,\text{V}$ (ample margin over the 14.3 V OV trip), $R_{ON} = 2.4\,\text{m}\Omega$ typ (negligible conduction loss at this design's currents), $V_{to} = 2.9\,\text{V}$ (comfortably below the LTC4215's 4.7–6.5 V gate drive).

Resolved: Si7336ADP confirmed as the final BOM part (not the datasheet's original FDC653N) – reusing a simulated-and-laid-out part beats re-validating a new one for this project's scope. See Section 6 for a footprint/symbol pin-mapping issue this choice ran into.

4.2.4 TVS substitute for simulation

The real BOM part (P6KE16A, a power TVS) has no local SPICE model. **BZX84C15L** (OnSemi, 15 V zener) is used as a simulation-only stand-in for the clamp voltage – it reproduces the clamping I-V behavior at low test currents but is *not* power-rated for a real transient-suppression event. The real BOM still specifies P6KE16A.

5 SPICE Simulation (LTspice)

5.1 Setup

Supply, V_1	step/PWL (not flat DC) – see Section 5.2
Output load	none (bare $C_L = 12\,\text{mF}$)

Numbers below come from the `.raw` file (parsed with `numpy`, time as float64 + each variable as float32 per the LTspice binary layout, not read visually from plots).

5.2 Two debugging pitfalls, documented for reuse

Pitfall 1 – flat DC source masks the entire power-up transient. The first simulation attempt used $V_1 = 12\,\text{V}$ (flat DC). LTspice computes the transient's initial state from the circuit's own DC operating point, which for a flat supply is self-consistent at "board already powered, gate already decided its state" – with nothing left to change over time, the solver (reached via Gmin stepping; direct Newton failed) converged on an all-zero state and the 100 ms run showed $V_{OUT} = 0$ for its entire length. **Fix:** represent the connector-insertion event explicitly with a step source, `PULSE(0 12 0 1u 1u 1 1)`.

Pitfall 2 – the 100 ms internal startup delay is easy to mistake for a dead circuit. Even after Pitfall 1 was fixed, V_{OUT} still stayed at 0 V for the full 100 ms run. Tracing internal nodes (UV, OV, Gate, SS, INTV$_{CC}$/TIMER) showed every comparator settling at its expected, correctly-calculated value, and the ON pin's INTV$_{CC}$-referenced rising edge occurring normally around $t \approx 49\,\mu\text{s}$ – but the gate never moved afterward. The cause was the documented ON-tied-high behavior (Section 3): the FET-On bit does not go high until 100 ms *after power-up*, not on the ON pin's own edge. Extending the stop time to 200 ms reproduced the full, correct startup sequence. **Lesson: with TIMER and ON both tied to INTV$_{CC}$, any `.tran` stop time under ≈150 ms will show a circuit that looks dead but isn't.**

5.3 Results

5.3.1 Controlled startup

Gate drive begins	$t \approx 100.9\,\text{ms}$ (immediately after the internal delay)
SS ramp starts	$t \approx 105\,\text{ms}$
Inrush current peak	5.2 A – matches the 5 A circuit-breaker design target
V_{OUT} reaches 12 V	$t \approx 152\,\text{ms}$
Charge time ($C_L = 12\,\text{mF}$)	$\approx 51\,\text{ms}$

The inrush current peak landing almost exactly at the 5 A circuit-breaker threshold (rather than well above or below it) is a direct consequence of the foldback current-limit curve converging with the fixed circuit-breaker threshold as FB approaches its good level – not a coincidence, and a useful numerical cross-check that the SS/R_{SENSE} combination is sized consistently with the breaker.

5.3.2 Undervoltage (UV) fault and automatic recovery

Test: V_{DD} stepped from 12 V to 8 V at $t = 300\,\text{ms}$ (PWL), held for 20 ms, returned to 12 V at $t = 320\,\text{ms}$.

V_{OUT} sags to $\approx V_{DD}$	within $\approx 300\,\mu\text{s}$ of the UV step
Mechanism	M_1's body diode conducting $V_{OUT} \rightarrow V_{DD}$ (not an active pull-down)
V_{DD} recovers	$t = 320\,\text{ms}$
Gate re-enables	$t \approx 426$–$432\,\text{ms}$ (≈ 106–$112\,\text{ms}$ after V_{DD} recovery)
V_{OUT} back at 12 V	$t \approx 477\,\text{ms}$

Two findings worth separating: (1) the fast ($\approx 300\,\mu\text{s}$) output sag during the UV event is *not* the LTC4215 actively protecting anything – it is the unavoidable consequence of using a single N-channel series MOSFET as a high-side switch: once the gate turns off, the intrinsic body diode still passes current from the (higher) output back to the (lower) input. A design needing true reverse blocking would need back-to-back MOSFETs, not a single device. (2) The recovery delay (≈ 106–$112\,\text{ms}$) confirms the same internal 100 ms startup-sequencer behavior from Section 5.2 re-arms on every restart, not just the very first power-up – auto-retry ($uvautoretry = 1$) is fully automatic, but not instantaneous.

5.3.3 Overcurrent (OC) fault and permanent latch-off

Test: a voltage-controlled switch ($R_{ON} = 10\,\text{m}\Omega$) closed across V_{OUT} at $t = 550\,\text{ms}$ (after the UV test had already fully recovered), modeling a hard downstream short.

Total fault-path resistance	$R_{SENSE} + R_{ON,sw} + R_{DS(on)} \approx 17.4\,\text{m}\Omega$
Theoretical instantaneous peak	$V_{OUT}/R \approx 12\,\text{V}/17.4\,\text{m}\Omega \approx 690\,\text{A}$
Breaker response	trips within $t_{D(OC)} = 20\,\mu\text{s}$ (datasheet filter)
V_{OUT} after trip	0 V, held through the rest of the run (no recovery)

The very low total fault-path resistance is exactly why the 20 µs circuit-breaker filter matters: without a fast-acting, independent breaker comparator, this near-ideal short-circuit path could sustain hundreds of amps rather than the 5–15 A the design targets. The permanent latch ($ocautoretry = 0$, no recovery for the remainder of the 700 ms run) is the intended behavior and the clearest contrast with the UV case above: a real overcurrent fault does not get a second chance without an external reset.

6 PCB Layout (KiCad)

6.1 Library sourcing

U1: symbol, footprint, and STEP model for `LTC4215IUFD-1_PBF` sourced from SnapEDA (via DigiKey). Its own `MPN` metadata field incorrectly read `LTC4215IUFD-2#TRPBF` (a data-entry error on SnapEDA's part, inconsistent with every other field on the same symbol) – cross-checked against the part's real datasheet (`421513fc.pdf`) before trusting it; the 24 QFN-24 pin assignments were also independently re-verified against the datasheet's own pin configuration diagram before use, not assumed correct because the source was reputable.

Q1: reused the `Transistor_FET:Si7336ADP` symbol and `Package_SO: PowerPAK_SO-8_Single` footprint, both already validated in `VRM_2PHASE_LTC1929`. **Pitfall found and fixed:** the generic `Q_NMOS_GSD` symbol (3 pins: 1=G, 2=S, 3=D) does *not* match this footprint's real pad numbering (pads 1–3 = Source, pad 4 = Gate, pad 5 ×5 = Drain) – using the generic symbol silently routed the Drain net onto a physical Source pad and left two pads unconnected. The dedicated `Si7336ADP` symbol (pins 1–3=S, 4=G, 5=D) matches the footprint exactly and resolves this with no footprint editing required.

6.2 Assembly: SMD except where physically impossible

All-SMD assembly, since U1 (QFN-24) and Q1 (PowerPAK SO-8) are SMD-only parts regardless – going THT for the passives would not simplify assembly and would only grow the board. Three exceptions, each because no SMD option exists at the required value/rating:

Part	Footprint	Why THT
D_1 (P6KE16A)	`Diode_THT:D_DO-15_P10.16mm_Horizontal`	no SMD option at 600 W peak pul
C_4 (12 mF)	`Capacitor_THT:CP_Radial_D16.0mm_P7.50mm`	no ceramic SMD exists past tens o
J_1, J_2	`Connector_AMASS:AMASS_XT60-M_1x02_P7.20mm_Vertical`	connector, not a 2-pad part

C_4 **real part:** Chemi-Con `EKYA160ELL123MLP1S`, 12,000 µF/16 V (margin over the 14.3 V OV trip), radial leaded, 16 mm⌀ × 37 mm, 7.5 mm pitch.

Connector choice, XT30 → XT60: XT30 (rated ≈ 30 A, better matched to this design's 15 A steady-state limit than the 60 A-rated XT60) was the first choice, but no XT30 STEP model exists anywhere in the local KiCad 3D-model package (confirmed by a full filesystem search), while XT60's does. Since the project is not being physically built, the connector's only remaining purpose is to communicate the hot-swap use case in a 3D render – switched to XT60 (more current margin than needed, but renders correctly) rather than spend effort sourcing an XT30 model that would not change anything else about the design.

R_{SENSE} **(R5):** `Resistor_SMD:R_2512_6332Metric`, not 0805 – needs the larger package both for power dissipation at 5 mΩ and because the datasheet's own Figure 5 requires a Kelvin connection at this part, which a 2-terminal 0805 cannot support as cleanly.

6.3 Layout, per Datasheet Figure 5

- **UV/OV/FB dividers kept close to U1**, short traces to V_{DD}/GND – Figure 5's own recommended layout shows exactly this. An early placement pass put the FB divider (R_7/R_8) near the output connector instead (logical by net name, wrong by the datasheet's own guidance) and was relocated next to U1.

- **Kelvin connection at R_{SENSE}**: SENSE+ (tied to the same trace as V_{DD}, per the datasheet – no separate tap needed on that side) and SENSE− routed as a dedicated, low-current tap branching directly off R_{SENSE}'s own pad, not off the high-current trace further downstream toward Q1 – any IR drop in a shared segment would otherwise corrupt the current reading.
- **2-layer stackup**: bottom layer solid GND pour (single ground domain, 15 A doesn't justify 4 layers, same reasoning as `BUCK_LM259`); top layer carries signal/power routing plus dedicated +12V and SOURCE/V_{OUT} copper pours (not just traces) sized for the 15 A steady-state current, per IPC-2221 at 1 oz copper (≈11 mm equivalent trace width).
- **Via stitching**: multiple vias in parallel (not one larger via – a single standard via only carries ≈1–2 A) under U1's exposed pad (EP, pin 25) and at the high-current GND return points near D_1/the connectors, same technique as the 3×3 thermal-pad via grid in `BUCK_LM259`. Stitching vias placed adjacent to pads, not via-in-pad, to avoid the solder-wicking risk already identified in that project.
- **Thermal vias under R5 and Q1** (added after the thermal FEM, Section 7.4): 9 on SENSE− landing on a SENSE− island cut out of the B.Cu GND pour, and 2 on +12V. Some of these are via-in-pad, an exception to the previous rule that would need tented/plugged vias in a real build.

6.4 DRC: two more lessons, same pattern as Section 5.2

QFN-24 copper clearance. DRC flagged 7 clearance violations, all between adjacent U1 pads: the board's `Default` net class requires 0.2 mm clearance, but this 0.5 mm-pitch QFN's real pad-to-pad spacing is 0.1778 mm – a rule mismatch, not a layout error. Fixed by lowering the `Default` net class clearance to 0.15 mm (still a standard, broadly fabricable value), not by editing the footprint.

Solder mask aperture bridging. A second DRC pass then flagged 18 "front solder mask aperture bridges items with different nets" errors, also all on U1: even with solder mask expansion already at 0 mm, adjacent pads on a 0.5 mm pitch QFN are close enough that their mask openings merge into one. This is normal and fabricator-accepted for fine-pitch parts (the paste stencil is designed separately); the fix is the `Allow bridged solder mask apertures between pads within footprints` checkbox in Board Setup → Solder Mask/Paste, not a geometry change.

Final status: ERC and DRC both clean.

7 Thermal FEM (COMSOL)

7.1 Method

Same criterion as the LM317 and LM2596 projects: COMSOL Multiphysics 6.4, *Heat Transfer in Solids*, natural convection to 25 °C still air, no heatsink, **no radiation** (conservative; added only as a variant). Steady state at the design's continuous operating point: 12 V, 15 A (the post-startup steady-state current limit, not the 5 A startup breaker, which is a transient and irrelevant to steady-state temperature). Unlike the two earlier projects, the heating of the copper itself is *computed*, not neglected: an *Electric Currents, Shell* model solves how the 15 A actually spreads through the pours, coupled both ways with the thermal model. The whole model is built and solved from a script (`fem/codigo/termico_efuse.py`), so it can be regenerated; the solved model is `fem/EFUSE_termico.mph`.

- **Geometry.** STEP exported with `kicad-cli pcb export step --subst-models` (`fem/geometria/EFUSE.step`), board and component bodies only – *not* with `--include-tracks/pads/zones`. That option was deliberately not used: in the LM2596 project the 35 µm copper solids it produces could not be meshed within the machine's memory (the mesher failed on tiny features, and a finer mesh was killed by the OS at 11.5 GB). Imported automatically into COMSOL with *Form Union* (no manual contact pairs needed).
- **Copper as a thin layer with the real layout.** F.Cu and B.Cu are exported as SVG, rasterized at 0.1 mm into copper-fraction maps (`fem/codigo/mascaras_cobre.py` → `fem/geometria/mascara_cobre_*.txt`; F.Cu 75 % coverage – the +12 V, SENSE− and SOURCE pours – and B.Cu 90 % – the GND pour plus the SENSE− island described below), and applied as conductive *Thin Layers* of 35 µm on the board's top and bottom faces, $k = k_{FR4} + (k_{Cu} - k_{FR4})\, m(x, y)$. Every pour, trace and clearance conducts exactly where it is on the board.
- **Solder.** KiCad's 3D models start at $z = 1.595$ mm and the board at 1.51 mm: without filling that 85 µm gap the components are thermally disconnected. SAC305 blocks were added under R5's two terminals, Q1's drain pad and four leads, and U1's exposed pad.
- **Vias.** Read directly from the `.kicad_pcb` (position and drill) and modeled as solid cylinders with the equivalent conductivity of a 25 µm copper barrel in a 0.3 mm drill (122 $\mathrm{W\,m^{-1}\,K^{-1}}$). The current board has 27: 16 GND stitching vias around U1, plus **11 thermal vias added after the first FEM pass** (Section 7.4).
- **Materials.** FR4 $k = 0.3$, copper 400, SAC305 58, R5 body as alumina 25, Q1/U1 molding compound (EMC) 0.7, other parts generic filler 0.2 ($\mathrm{W\,m^{-1}\,K^{-1}}$). The STEP models of Q1 and U1 include their drain pad / exposed pad as separate solids, modeled as copper.
- **Convection.** One correlation per face orientation (horizontal plate up $Nu = 0.54\, Ra^{1/4}$, down $Nu = 0.27\, Ra^{1/4}$, $L = 7.7$ mm = board area/perimeter; vertical wall Churchill–Chu, $L = 5$ mm), written explicitly because COMSOL's built-in natural-convection correlations returned NaN in the LM2596 model.
- **Mesh.** Free tetrahedra with the geometry stretched 10× in z while meshing: 285,659 elements. Solved under a 12 GB memory cap.

7.2 Copper I^2R heating (coupled electrical–thermal)

The 15 A follows four separate copper paths, each solved on the same thin layers as the heat:

Net	Layer	Current enters at	Leaves at
+12V	F.Cu	J1 pin 1	R5 pad 2
SENSE−	F.Cu	R5 pad 1	Q1 drain pad
SOURCE	F.Cu	Q1 pins 1–3	J2 pin 1
GND (return)	B.Cu	J2 pin 2	J1 pin 2

- Conductivity $\sigma = \sigma_{Cu}(T)\, m(x, y)$ with the same copper maps, so the current can only flow where there is copper; $\sigma_{Cu}(T) = 5.96 \times 10^7\,\mathrm{S\,m^{-1}}/[1 + 0.00393\,(T - 20\,^\circ\mathrm{C})]$, so hot copper has more resistance and dissipates more (two-way coupling, solved together).
- The current is injected as a source spread over the input pad's copper (normalized to exactly 15 A); the output pad is the 0 V reference. The Joule heat $\sigma|E|^2 t$ is fed back as a heat source on the same faces.
- **Pitfall found:** in COMSOL 6.4 the shell interface ignored the thickness given to it and used its default (0.1 mm instead of 35 µm), which would have underestimated the loss by 3×. Worked

around by scaling σ so the sheet conductance σt is correct, and checked against a strip with an analytic solution ($P = I^2RL/3$ for uniform injection): exact match.
- Simplifications: the vias do not carry current between layers in the electrical model (all of SENSE−'s current stays in F.Cu – conservative), and the THT connector pins inject on one layer only.

7.3 Heat sources at 15 A

Source	Loss	Origin	Where it enters the model
R5 (R_{SENSE}, 5 mΩ, 2512)	1.125 W	$I^2R = 15^2 \times 0.005$	volume of the resistor
Q1 (Si7336ADP, $R_{DS(on)} = 2.4\,\text{m}\Omega$ typ)	0.54 W	$I^2R_{DS(on)}$, conduction only	face between drain pad and die
U1 (LTC4215)	0.036 W	$V_{DD} \times I_{DD}$ (3 mA typ)	face of the exposed pad
F.Cu pours (+12V, SENSE−, SOURCE)	1.03 W	computed, Section 7.2	where the current flows
B.Cu GND return	0.58 W	computed, Section 7.2	where the current flows
Total	3.31 W		

The copper itself dissipates 1.61 W **– as much as R5, Q1 and U1 together (**1.70 W**)**, nearly doubling the heat on the board. The earlier hand estimate (≈0.9 W) was low by almost 2×: the current crowds into the narrow parts of the pours, and the copper is at 100 °C to 160 °C, where its resistance is 30 % to 55 % higher than at room temperature. Q1 has no switching loss: it only ramps once at startup and then sits fully enhanced. Its $R_{DS(on)}$ is the 25 °C typical value; since $R_{DS(on)}$ rises with temperature, a variant with $1.5 \times R_{DS(on)}$ (0.81 W) is included below.

7.4 Thermal vias under R5 and Q1

The first FEM pass (no copper heating yet) showed that the 16 original vias all stitch U1's ground and none reach the two hottest parts. Since B.Cu is a GND pour and R5/Q1 sit on +12V and SENSE−, vias cannot go straight down (they would short to GND): a copper island of the *same* net has to be cut out of the GND pour first. Added on the board:
- **9 vias on SENSE−**: in and around Q1's drain pad and next to R5's SENSE− terminal, landing on a SENSE− island cut out of the B.Cu GND pour (visible in Figure 3).
- **2 vias on +12V** in R5's other terminal. DRC flags them as *dangling*: there is no +12V copper on B.Cu, so they only conduct through their own barrel and barely help. A +12V island on B.Cu under R5 would make them useful.

Several of these vias are inside the pads (via-in-pad), which the layout section had avoided because of solder wicking: for a real build they should be tented/plugged, or moved to the pad edge.

7.5 Results at 12 V, 15 A

The three columns separate the two effects: what the copper heating adds (first → second) and what the new vias take away (second → third).

	First pass (no copper heating, no vias)	Copper heating, original board	**Copper heating, with thermal vias**
Convection only (design case):			
R5 max	139.1 °C	178.9 °C	170.2 °C
Q1 junction (die face)	118.7 °C	158.7 °C	148.9 °C
U1 max	87.1 °C	127.2 °C	126.8 °C
Board max	127.8 °C	168.0 °C	158.0 °C
with $R_{DS(on)} \times 1.5$: R5 / Q1	150.6 / 135.9 °C	190.8 / 176.1 °C	180.7 / 163.9 °C
+ radiation ($\varepsilon = 0.9$, solder mask):			
R5 max	117.3 °C	141.5 °C	133.3 °C
Q1 junction	97.8 °C	122.0 °C	112.3 °C
U1 max	67.8 °C	91.7 °C	91.3 °C
with $R_{DS(on)} \times 1.5$: R5 / Q1	126.0 / 112.2 °C	150.0 / 136.0 °C	140.6 / 124.1 °C
Copper I^2R (F.Cu + B.Cu)	not modeled	1.60 W	1.61 W

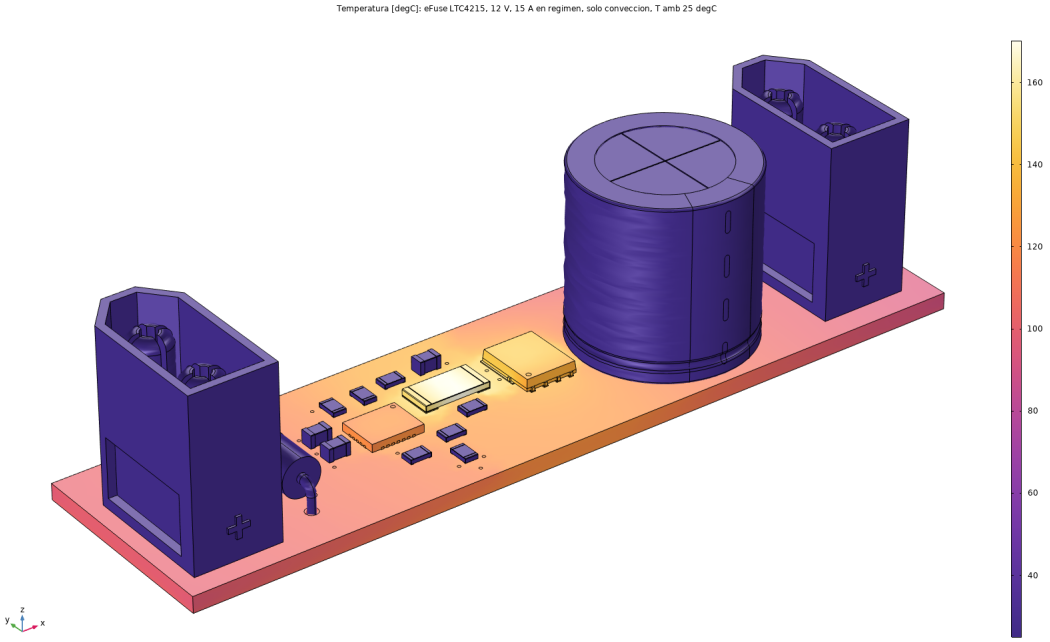


Figure 1: Temperature at 12 V, 15 A, convection only, copper heating and thermal vias included. R5 (center) is the hottest part, followed by Q1 (right of R5); the whole power path is hot because the pours themselves dissipate 1.6 W.

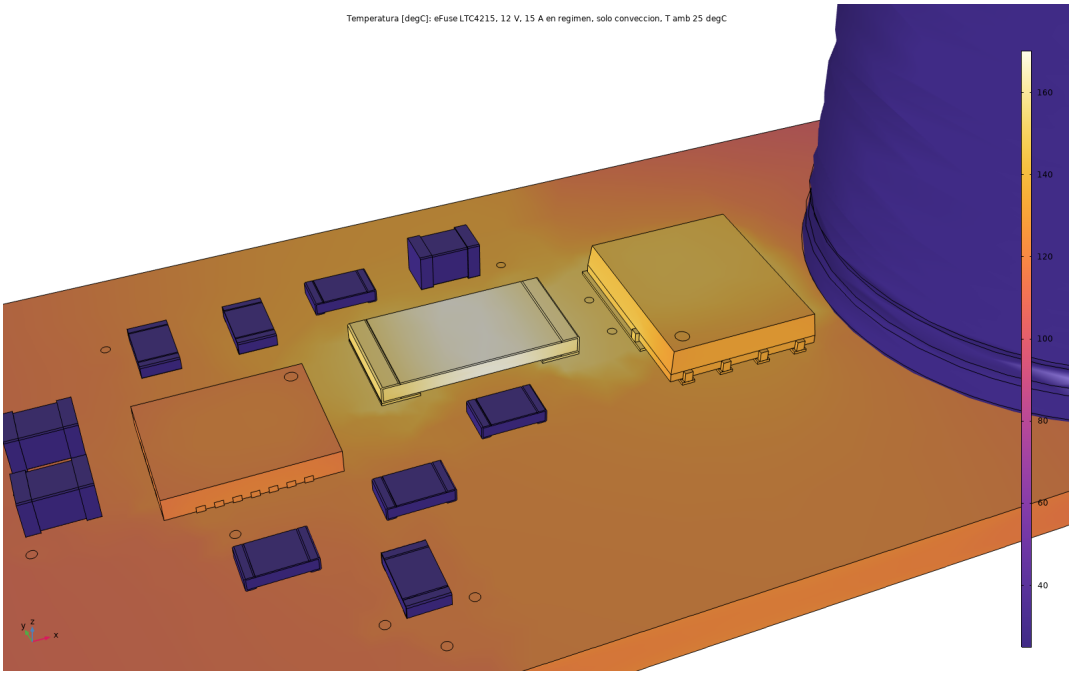


Figure 2: Close-up of R5 and Q1.

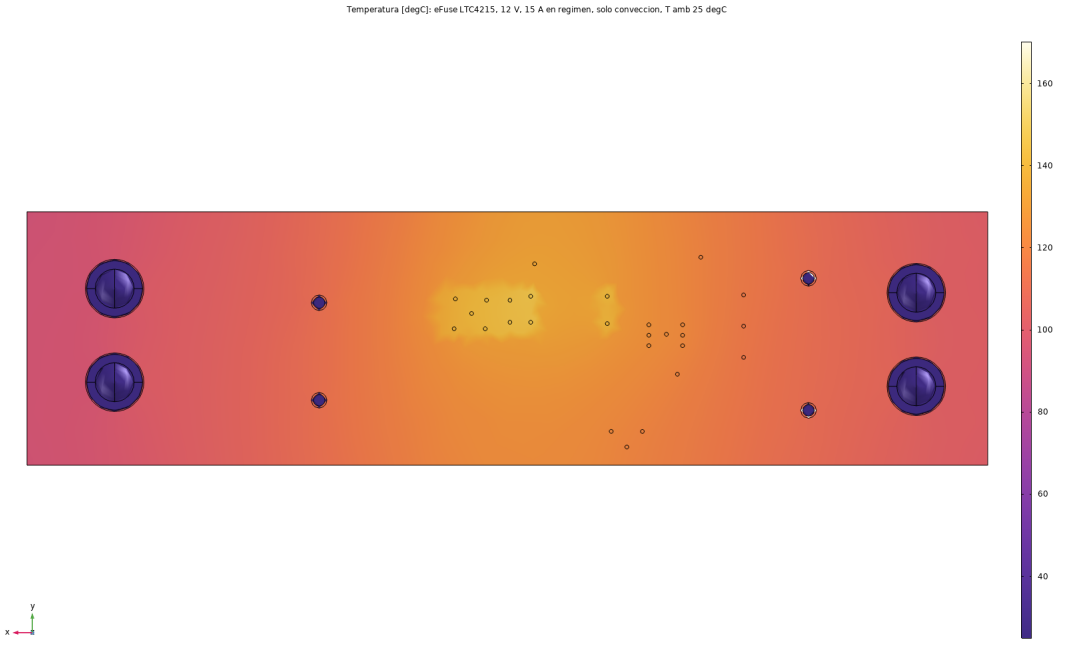


Figure 3: Bottom side (seen from below, so mirrored left–right). The warmest spots are the new SENSE− island with its vias under R5/Q1 and the two +12V vias: the vias carry heat to the bottom layer.

Current sweep (final board, convection only, $R_{DS(on)}$ at 25 °C). P is R5 + Q1 + U1 only; the copper adds roughly as much again.

I	P (parts)	R5 max	Q1 junction	U1 max
5 A	0.22 W	45.7 °C	43.2 °C	41.8 °C
7.5 A	0.45 W	66.1 °C	60.6 °C	56.1 °C
10 A	0.78 W	93.5 °C	83.9 °C	74.8 °C
12.5 A	1.19 W	127.9 °C	113.0 °C	98.2 °C
15 A	1.70 W	170.2 °C	148.9 °C	126.8 °C

7.6 Findings

- **The copper heating is the dominant missing piece**: 1.6 W, adding 30–40 °C to every part. The first pass, without it, was optimistic.
- **The thermal vias work, but the gain is modest**: −9 °C on R5 and −10 °C on Q1, in every variant. They help move heat to B.Cu, but the problem is now the total heat on a small board, not only how it leaves R5 and Q1.
- **Continuous 15 A is not thermally safe in still air.** With convection only, Q1's junction sits right at its 150 °C $T_{J,max}$ (164 °C with a hot $R_{DS(on)}$), and R5 reaches 170 °C, at or above typical 2512 maximum operating temperatures (155–170 °C). Counting radiation (realistic for a solder-masked board) it is 112–124 °C on Q1 and 133–141 °C on R5: within limits but with little margin.
- **R5, not the MOSFET, is still the hottest component** – same pattern as the LM2596 project, where the catch diode, not the regulator IC, limited the design.
- 12.5 A **continuous is comfortable**: 128 °C on R5 and 113 °C on Q1 even with convection only.
- **To reach** 15 A **continuous**, in order of effect: 2 oz copper (halves the 1.6 W copper loss), wider/shorter power pours, a metal-strip R5 (≥2 W, higher temperature rating), any airflow, and a +12V island on B.Cu so the two dangling vias work.

Not modeled. The XT60 connectors and the large capacitor sit 85 µm above the board in the STEP with no solder under them, so they are thermally floating; in reality the connectors and the input/output cables act as heat sinks (and as extra I^2R sources at their contacts). Vias carry heat but not current between layers in the electrical model.

8 Known Limitations

- **No reverse blocking.** As shown in Section 5.3, a single N-channel series MOSFET's body diode will always pass current from output to input if the output is briefly higher than the input, independent of the LTC4215's own gate control. Acceptable for this project's scope (inrush/OC protection into a load), not acceptable where true input disconnection is required.
- **Simulation-only substitute part.** The TVS (BZX84C15L in place of P6KE16A) is a SPICE-availability substitute only – the real BOM/PCB correctly specifies P6KE16A (Section 4.2.4, Section 6).
- **No I²C/ADC validation.** Monitoring, alerting, and the 8-bit ADC are real LTC4215 features left entirely untested here (Section 4.2.1).

- **Thermal: continuous 15 A not safe in still air.** With the copper's own 1.6 W of I^2R included, COMSOL gives 170 °C on R5 and 149 °C on Q1's junction (convection only, with the thermal vias); 12.5 A continuous is comfortable (Section 7).
- **Silkscreen overlap.** DRC reports cosmetic silkscreen-clearance warnings around the densely packed U1/divider area. Not addressed – irrelevant with no physical build planned (Section 6).

9 Pending

PENDING

- Physical build/measurement – out of scope for this project (simulation + layout only, confirmed at project start), unlike `RLC_PID_UNI`.
- Follow-on STM32 + I^2C telemetry project (Section 4.2.1).
- For continuous 15 A: 2 oz copper and/or wider pours, a specific metal-strip R5, and a +12V island on B.Cu for the two dangling vias – then re-run the FEM (the script regenerates everything from the `.kicad_pcb`).

10 Conclusions

A 12 V hot-swap/eFuse front-end built around the LTC4215 was designed from the manufacturer's own reference circuit, with every UV/OV/FB threshold independently re-derived and numerically cross-checked against the datasheet's comparator thresholds rather than copied blindly. LTspice simulation confirmed all three headline behaviors of the design: a soft-start-controlled inrush whose peak current lands on the calculated 5 A circuit-breaker target, fully automatic recovery from an undervoltage event (with a real, non-obvious ≈ 100 ms restart delay), and a permanent, non-resettable latch-off on a hard output short. Two simulation pitfalls specific to this IC's behavior (a flat-DC source hiding the power-up transient, and the internal 100 ms startup-sequencer delay) are documented in Section 5.2 so they are not re-discovered from scratch on a future project using the same IC family.

The KiCad schematic and 2-layer PCB layout carry the design through to a fabricable, ERC/DRC-clean board: a real, independently-verified QFN-24 part (not blindly trusted from its distributor library), a corrected MOSFET footprint/symbol pin mapping, a Kelvin-connected sense resistor, and dedicated copper pours sized for the 15 A steady-state current. Two more IC/package-specific gotchas (fine-pitch QFN copper clearance and solder-mask aperture bridging, Section 6) are documented alongside the two simulation pitfalls, for the same reason.

A COMSOL steady-state model at the continuous 12 V/15 A point, with the current flow in the copper solved together with the heat (Section 7), found that **the pours themselves dissipate** 1.6 W, as much as all the components together, and that the **sense resistor R5**, not the MOSFET, is the hottest part. Thermal vias under R5 and Q1, added in response to the first pass, lower both by about 10 °C, but in still air continuous 15 A still puts R5 at 170 °C and Q1's junction at its 150 °C limit; 12.5 A continuous is comfortable, and 15 A needs heavier copper, a metal-strip R5 or airflow.